

Student's Name:.....

Teacher's Name:.....



Year 11

Half Yearly Examination

2010

MATHEMATICS ADVANCED

Time allowed – 2 hours
(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES:

- Attempt ALL questions. Marks shown are a guide only.
- All necessary working must be shown.
- Write in blue or black pen and do not use liquid paper
- Write your teacher's name and your name on this cover sheet.

Question 1. Start on a new page

- a) Evaluate correct to two significant figures: $\sqrt[3]{\frac{5.21+4.71}{3.94 \times 1.76}}$ 2
- b) Factorise fully : $10x + 2xy - 10y - 2y^2$ 2
- c) Simplify : $2\sqrt{3} - \sqrt{27} + \sqrt{48}$ 2
- d) Expand and simplify: $(x-3)(x^2 + 5x - 1)$ 2
- e) If $f(x) = 9 - 2x^2$, find the value of $f(-1)$ 1

Question 2. Start on a new page

- a) Rationalise the denominator : $\frac{5}{5-\sqrt{2}}$ 2
- b) If $T = 2\pi\sqrt{\frac{l}{g}}$
- i) Find the value of T, if $l = 80$ and $g = 9.8$ 1
- ii) Make g the subject of the formula 2
- c) Solve for x: $\frac{3x}{x+1} - \frac{2}{3} = 0$ 3
- d) Express 0.142 as a fraction in its simplest terms. 1

Question 3. Start on a new page

- a) Factorise completely
- i) $8x^3 - 1$ 2
- ii) $(2a - b + c)^2 - (a + 5b - c)^2$ 2
- b) Find the domain and the range of $y = \frac{3}{2x-1}$ 2
- c) Evaluate $\sin 60^\circ + \tan^2 30^\circ$. Leave your answer in exact form. 2

Question 4. Start on a new page

a) Solve

i. $3x + y = 4$

$2x + 3y = 1$

3

ii. $x^4 - 16 = 0$

2

iii. $\frac{1-8x}{5} < 9$

3

b) Find the value of a^3b^2 in the index form if $a = (\frac{1}{2})^3$ and $b = (\frac{4}{5})^2$, expressing your answer in the form of 2^x5^y

2

c) Simplify $\frac{1}{\sqrt{2}+1} + \frac{2}{\sqrt{2}-1}$

2

Question 5. Start on a new page

a) Sketch the following and show the important features. Write down the domain and range of each function.

i. $y = x^2 - x - 6$

3

ii. $y = (x-3)^3$

3

b) Solve $|2x + 1| \leq 7$ and graph the solution on a number line

3

Question 6. Start on a new page

a) Simplify $\frac{3}{x+1} + \frac{2}{x^2-1} - \frac{4}{x-1}$

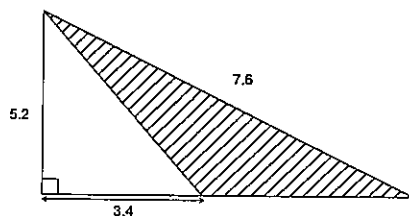
3

b) Solve $|5x - 1| = 3x + 3$

3

c) Find the area of the shaded region correct to two decimal places:

3



Question 7. Start on a new page

a) If $f(x) = \begin{cases} x^2 & \text{when } x \geq 2 \\ 5x & \text{when } 0 \leq x < 2 \\ -3 & \text{when } x < 0 \end{cases}$,

sketch the graph of $f(x)$ for $-3 \leq x \leq 5$

4

b) Find all possible values of A if $2 \sin A = \cos A$ and $0^\circ \leq A \leq 360^\circ$

2

c) Solve $9x^2 + 6x - 4 = 0$ by completing the square. Leave the answer in exact form

3

Question 8. Start on a new page

a) Three points A, B and C lie on a horizontal ground. From B the bearing of A is $047^\circ T$ and the bearing of C is $081^\circ T$. From A the bearing of C is $149^\circ T$. The distance between A and B is 10km.

i. Draw a diagram and mark all the information given above.

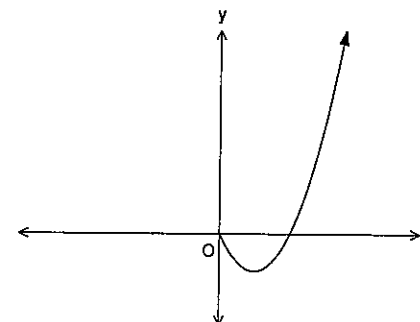
3

ii. Find the distance from A to C.

2

b) If $y = f(x)$ is an odd function, copy and complete the graph below

2



Question 9. Start on a new page

a) Find a value of β if $\cos \beta = \sin 43^\circ 50'$

2

b) i. Show that $\frac{x+3}{x+1}$ is $1 + \frac{2}{x+1}$

1

ii. Hence or otherwise sketch the graph of $y = \frac{x+3}{x+1}$ showing any intercepts and asymptotes

3

iii. If $f(x) = \frac{x^3 - 3x^2 + 1}{x(1-x)}$, prove that $f(\frac{1}{x}) = f(1-x)$ where $x \neq 0, 1$

3

END OF EXAMINATION

1. a) $1.1267 \dots \checkmark$
 $1.1 \checkmark$ (2)

b) $2(5x + xy - 5y - y^2)$
 $= 2(5+y)(x-y)$ (2)

c) $= 2\sqrt{3} - 3\sqrt{3} + 4\sqrt{3} \checkmark$
 $= 3\sqrt{3} \checkmark$ (2)

d) $= x(x^2 + 5x - 1) - 3(x^2 + 5x - 1) \checkmark$
 $= x^3 + 5x^2 - x - 3x^2 - 15x + 3$
 $= x^3 + 2x^2 - 16x + 3 \checkmark$ (2)

e) $f(-1) = 9 - 2(-1)^2$
 $= 7 \checkmark$ (1)

(9)

2. a) $= \frac{5}{5-\sqrt{2}} \times \frac{5+\sqrt{2}}{5+\sqrt{2}} \checkmark$
 $= \frac{25+5\sqrt{2}}{23} \checkmark$ (2)

b) (i) $T = 2\pi \sqrt{\frac{80}{9.8}} \checkmark = 17.95195 \dots$
 $= 17.95$ (2dp) (2)

(ii) $T^2 = 4\pi^2 \times \frac{L}{g} \checkmark$
 $g = \frac{4\pi^2 L}{T^2} \checkmark$ (2)

c) $\frac{3x}{x+1} = \frac{2}{3} \Rightarrow 9x = 2x+2$
 $7x = 2 \therefore x = \frac{2}{7} \checkmark$ (3)

d) $x = \frac{47}{330} \checkmark$ (1)

(8)

3 a) (i) $= (2x-1)(4x^2+2x+1)$ (2)
 (ii) $= [(2a-b+c) - (a+5b-c)][2a-b+c+a]$
 $= (a+4b+2c)(3a+4b)$ (2)

b) $D = \{x: x \in \mathbb{R}, x \neq \frac{1}{2}\} \checkmark$
 $R = \{y: y \in \mathbb{R}, y \neq 0\} \checkmark$ (2)

c) $= \frac{\sqrt{3}}{2} + \frac{1}{3} \checkmark$
 $= \frac{3\sqrt{3}+2}{6} \checkmark$ (2) (8)

4 a) (i) $3x+y=4$ — ①
 $2x+3y=1$ — ②

① $\times 3$ — ② $\Rightarrow 7x = 11 \checkmark$
 $\therefore x = \frac{11}{7} \checkmark$
 $\therefore y = 4 - 3 \times \frac{11}{7} = -\frac{5}{7} \checkmark$ (3)

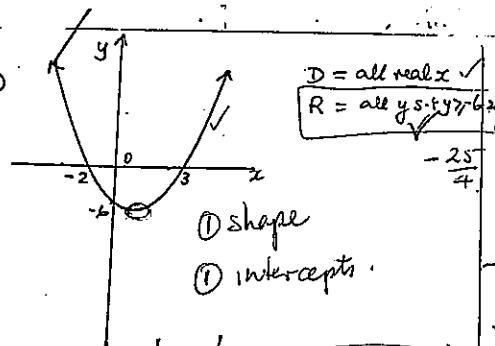
(ii) $(x^2-4)(x^2+4)=0$
 $(x-2)(x+2)(x^2+4)=0$
 $\therefore x=2, -2 \checkmark$ (2)

(iii) $\frac{1-8x}{5} < 9$
 $1-8x < 45 \checkmark$
 $-8x < 44$
 $8x > -44 \checkmark$
 $x > -\frac{11}{2} \checkmark$ (3)

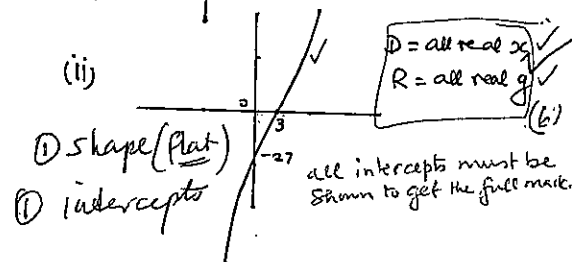
b) $a^3 b^2 = \left[\left(\frac{1}{2}\right)^3\right]^3 \left[\left(\frac{4}{5}\right)^2\right]^2 \checkmark$
 $= \frac{1}{2^9} \cdot \frac{2^{-8}}{5^{-4}} = \frac{1}{2^9} \cdot \frac{5^4}{2^8}$
 $= 2^{-17} 5^4 \checkmark$ (2)

c) $1(\sqrt{2}-1) + 2(\sqrt{2}+1) \checkmark$
 $= 3\sqrt{2}+1 \checkmark$ (2)

5 a) (i)



(ii)



b) $|2x+1| \leq 7$
 $2x+1 \leq 7 \quad - (2x+1) \leq 7$
 $x \leq 3 \checkmark \quad 2x+1 \geq -7$
 $-4 \leq x \leq 3$ needed to put $x \geq -4$ together (3)

6 a) $= \frac{3}{x+1} + \frac{2}{(x-1)(x+1)} - \frac{4}{(x-1)}$
 $= \frac{3(x-1) + 2 - 4(x+1)}{(x-1)(x+1)} \checkmark$
 $= \frac{-x-5}{x^2-1} \checkmark$ (3)

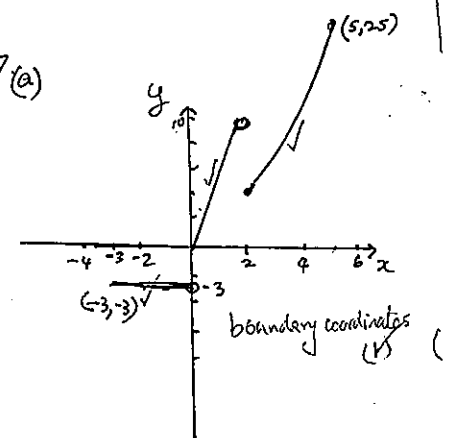
b) $|5x-1| = 3x+3$
 $5x-1 = 3x+3 \quad - (5x-1) = 3x+3$
 $2x = 4 \quad -8x = +2$
 $x = 2 \checkmark \quad x = -\frac{1}{4} \checkmark$

check: LHS = 9
 RHS = 9
 $\checkmark$

LHS = $\frac{9}{4}$
 RHS = $\frac{9}{4} \checkmark$ (3)

(c) Base = $\sqrt{7 \cdot 6^2 - 5 \cdot 2^2}$
 $= 5.542 \checkmark$
 Required area = $\frac{1}{2} \times 5.542 \times 5.2 - \frac{1 \cdot 3 \cdot 1}{2} \checkmark$
 $= 5.57$ (2 dp) (2)

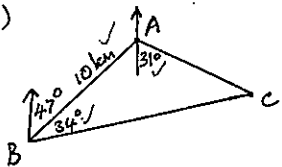
7(a)



b) $\tan A = \frac{1}{2}$
 $A = \tan^{-1}(\frac{1}{2}) \checkmark$
 $A = 26^\circ 34', 206^\circ 34'$ (2)

(c) $x^2 + \frac{6}{9}x - \frac{4}{9} = 0$
 $x^2 + \frac{6}{9}x + \left(\frac{3}{9}\right)^2 = \frac{4}{9} + \left(\frac{3}{9}\right)^2$
 $\left(x + \frac{3}{9}\right)^2 = \frac{5}{9}$
 $x = \pm \frac{\sqrt{5}}{3} - \frac{1}{3}$
 $x = \frac{(\sqrt{3}-1)}{3} \text{ or } -\frac{(\sqrt{3}+1)}{3} \checkmark$ (2)

8) a) (i)



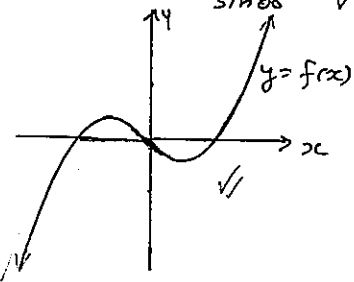
(3)

$$i) \angle BCA = 180 - (34 + 31 + 47) = 68^\circ$$

$$\frac{10}{\sin 68^\circ} = \frac{x}{\sin 34^\circ} \checkmark$$

$$x = 10 \times \frac{\sin 34^\circ}{\sin 68^\circ} = 6.03 \text{ km} \quad (2)$$

b)



(2)

(7)

9(a) $\cos \beta = \sin 43^\circ 50'$

$$\cos (90 - \alpha) = \sin \alpha$$

$$\therefore 90 - \alpha = \beta \checkmark$$

$$\beta = 90 - 43^\circ 50'$$

$$= 46^\circ 10' \checkmark$$

(2)

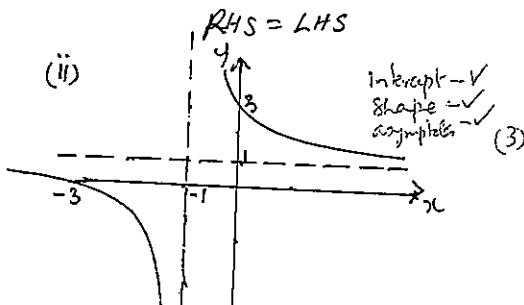
b) $RHS = 1 + \frac{2}{x+1} = \frac{x+1+2}{x+1} \checkmark$

(1)

$$= \frac{x+3}{x+1} \quad (1)$$

(1)

(ii)



(3)

iii) $f(x) = \frac{x^3 - 3x^2 + 1}{x(x-1)}$

$$f\left(\frac{1}{x}\right) = \frac{\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 + 1}{\frac{1}{x}\left(\frac{1}{x} - 1\right)} \checkmark$$

$$= \frac{1 - 3x + x^3}{x^2 x \left(\frac{x-1}{x}\right)} \checkmark$$

$$= \frac{1 - 3x + x^3}{x(x-1)} \checkmark$$

$$f(1-x) = \frac{(1-x)^3 - 3(1-x)^2 + 1}{(1-x)x}$$

$$= \frac{1 - 3x + 3x^2 - x^3 - 3 + 6x - 3x^2 + 1}{(1-x)x}$$

$$= \frac{-x^3 + 3x - 1}{(1-x)x}$$

$$= -\frac{(x^3 - 3x + 1)}{x(1-x)}$$

$$= \frac{x^3 - 3x + 1}{x(x-1)} \checkmark$$

(3)

Hence $f\left(\frac{1}{x}\right) = f(1-x)$.

(4)

